A Data-Driven Analysis and Prediction of Student Performance

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Abstract. This report examines student academic performance using exploratory analysis, regression models, and classification across the Mathematics and Portuguese datasets. The goal is to understand which factors influences performance and how well these can be used for further predictions.
From the exploratory analysis, past class failures stand out as the strongest factor. While there is a clear negative relationship with performance ($r = -0.37$ in Mathematics and $r = -0.41$ in Portuguese). Parental education shows a moderate positive relationship of $r = 0.21$ and $r = 0.29$, and study time appears to have a smaller impact. Most other variables show little to no clear relationship in both datasets.
A multiple linear regression model based on these variables resulted in a low explanatory power ($R^2 = 0.11$). The predictions are mostly centered around the mean, which suggests that the model is underfitting and not capturing strong patterns in the data.
A logistic regression model was also tested, classifying students into *Good* and *Not Good*, this model achieved an accuracy of 70.31% (90 out of 128 correctly classified), but still struggled to clearly separate the two groups.
Overall, the analysis reveals some general patterns in student performance. However, the models show limited predictive ability, might be because of weak relationships between the variables in the dataset.

1 Introduction

Student academic performance is influenced by a range of factors, and understanding these can help explain why some students perform better than others. When looking into these relationships it can also support more informed decisions in educational settings.

In this study, student performance is analysed using two datasets covering Mathematics and Portuguese subjects. The goal is to identify which variables are most closely related to performance(Score), and to examine how well these can be used to predict outcomes using statistical and machine learning models.

The analysis is carried out in three main steps, first, the exploratory data analysis is used to examine how variables such as past class failures, parental education, and study time relate to the students scores. This gives an initial understanding of the patterns found in the data. Second, regression models are applied to quantify these relationships and evaluate how much of the variation in performance that can actually be explained. Finally, a classification model is used to group students into performance categories and evaluate how well the model can separate these groups.

The results point to a similar pattern as the earlier analysis. Past academic performance, especially previous failures, stands out as the strongest factor related to student outcomes.

Parental education and study time also play a role, but their impact is smaller. Overall, the patterns are there, but the relationships in the data are not strong enough to support highly accurate predictions.

2 Data Preprocessing

2.1 Data Inspection and Quality Assessment

The datasets were inspected to get an overview of their structure, variable types, and overall quality. In total, three datasets were used: one prediction sample and two main datasets containing student performance in Mathematics and Portuguese.

By looking at the datasets using `.info()` and `.head()`, it was confirmed that both the main datasets contains 31 variables and no missing values. This means that no imputation was needed before continuing the analysis.

2.2 Handling of Missing Values and Duplicates

A check for duplicate records was performed using `.duplicated().sum()`. The result showed that there were no duplicates in either of the two dataset, meaning the data could be used as it is.

2.3 Exploratory Summary of Numerical Variables

Descriptive statistics were explored using `.describe()` to understand the distribution of the numerical variables. Most variables appear to fall within reasonable ranges. However, the *absences* variable stands out, with values going up to 75 in the Mathematics dataset. This could indicate a skewed distribution, and suggests that a few extreme values might influence the model.

Categorical variables were reviewed using `.unique()` to check the different categories present. All values appear consistent, with no unexpected or inconsistent entries. Based on this, the dataset can be considered relatively clean at this stage.

2.4 Encoding of Categorical Variables

Categorical variables were converted to numerical values so they could be included in the models. For example, binary variables like *higher* were encoded as 0 and 1 to make it easier to work with. Without this step, the regression models would not be able to use these variables for any predictions.

2.5 Feature Engineering

A new feature: `parent_edu_avg`, was created by combining the average education levels of the mother(Medu) and father's(Fedu) education into a single value.

Other versions, like the maximum and minimum education levels, were also tested and compared, but the differences were small. Because of that the average was chosen and used as a simple and practical variable that represents both the educational level of the parents.

2.6 Feature Transformation for Visualization

The parental education variable was rounded into whole numbers and stored as `parent_edu_cat` for visualization purposes.

This removes intermediate values (like, 2.5), which are less intuitive since the original variables are based on categories measured in whole numbers ranging from 0 (no formal education) to 4 (higher education), so the ordering is preserved.

For the modelling step, `parent_edu_avg` was kept, while `parent_edu_cat` was used only for visualization.

2.7 Preparation of Prediction Dataset

A `StudentID` variable was created and added to the prediction dataset to identify each observation, this makes it easier to match the predicted scores with the corresponding students and keeps the results more visual and structured.

2.8 Consistency Across Datasets

The same preprocessing steps were used for both the main datasets(Mathematics and Portuguese) to keep things consistent. This helps when comparing results across Mathematics and Portuguese.

Relevant feature engineering steps were also applied to the prediction dataset, this makes sure it fits the format expected by the trained model during the prediction.

3 Data Analysis

3.1 Nursery Attendance and Academic Performance (Mathematics)

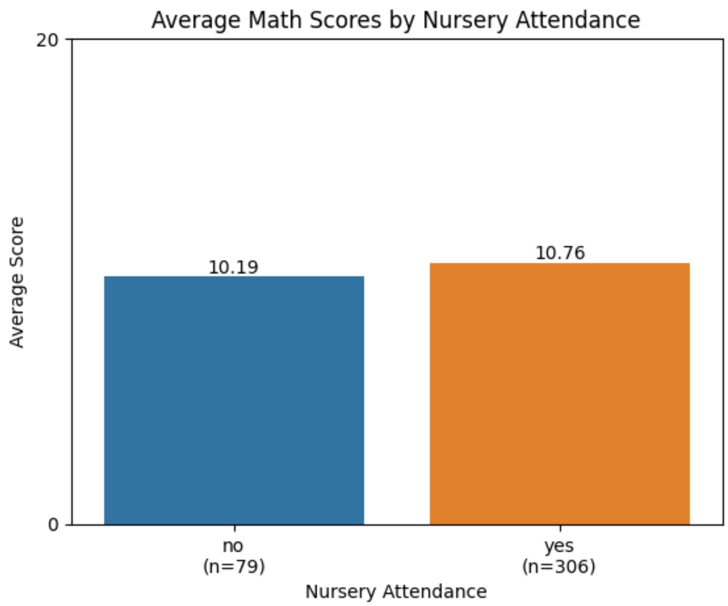

Fig. 1. Average Mathematics scores by nursery attendance. There is a small increase in scores for students who attended nursery school.

Figure 1, the average Mathematics scores are compared between students who attended nursery school and those who did not. Students who did not attend nursery school have an average score of 10.19 (n = 79), while students who attended have a slightly higher average of 10.76 (n = 306).

This gives a small difference of 0.57 points in favour of students who attended nursery school.

This could suggest a positive effect of early education, but the difference is relatively small. It therefore seems unlikely that nursery attendance alone has a strong impact on performance, although it may still contribute together with other factors.

It is also important to consider the difference in group sizes (79 vs 306 students), as the smaller group may be more sensitive to variations.

3.2 Nursery Attendance and Academic Performance (Portuguese)

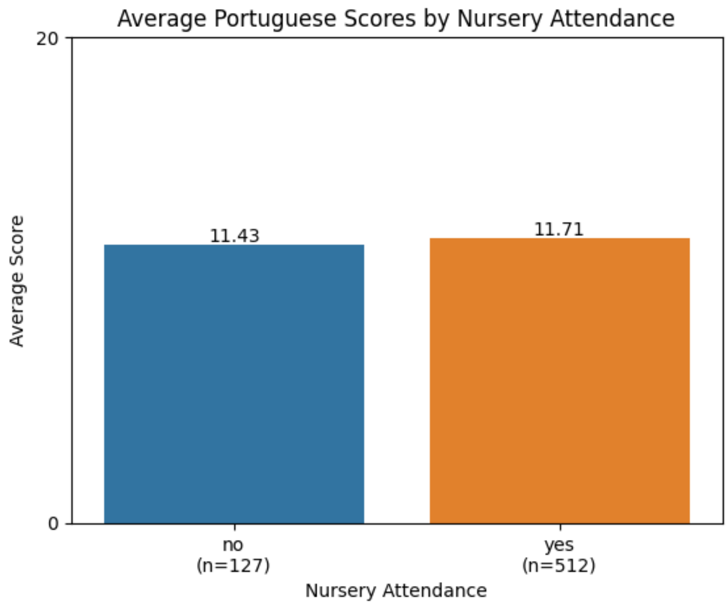

Fig. 2. Average Portuguese scores by nursery attendance. Students who attended nursery school have slightly higher scores.

In Figure 2, the average Portuguese scores are compared between students who attended nursery school and those who did not.

Students who attended nursery school have a slightly higher average score of 11.71 ($n = 512$), compared to 11.43 ($n = 127$) for students who did not attend. This gives a small difference of 0.28 points in favour of students with nursery attendance.

When comparing both subjects, a similar pattern appears, the difference in Mathematics was larger with 0.57 points, while in Portuguese it is smaller at 0.28 points. This could suggest that nursery attendance has a positive effect in both subjects, but the effect is relatively small.

As before, the group sizes are uneven, with many more students in the nursery group. This means the “no nursery” group is based on fewer observations and may be more sensitive to variation.

3.3 Parental Education and Nursery Attendance (Mathematics)

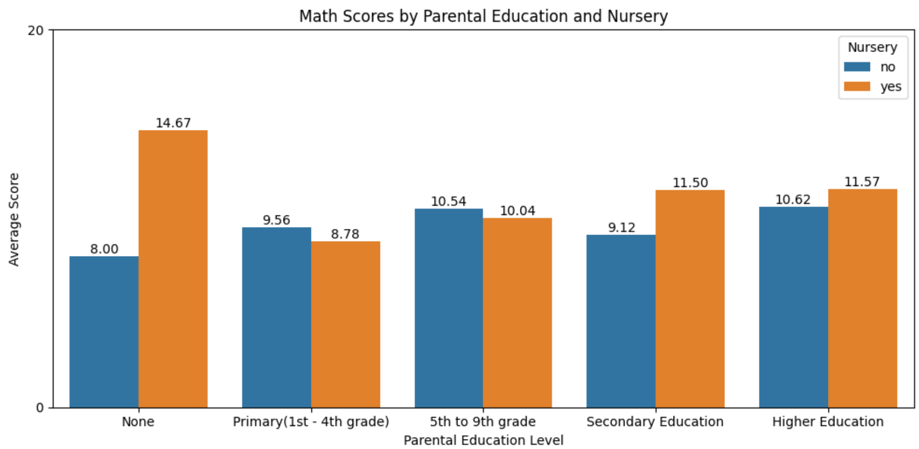

Fig. 3. Average Mathematics scores by parental education and nursery attendance. Students with nursery experience generally have higher scores.

Table 1. Average Mathematics Scores by Parental Education Level and Nursery Attendance with sample sizes

Nursery	**None**	**Primary**	**5th–9th**	**Secondary**	**Higher**
No	8.00 (n=1)	9.56 (n=15)	10.54 (n=40)	9.12 (n=8)	10.62 (n=15)
Yes	14.67 (n=1)	8.78 (n=24)	10.04 (n=118)	11.50 (n=55)	11.57 (n=108)

In Figure 3, the Mathematics scores are compared across different levels of parental education, while also distinguishing between students who attended nursery school and those who did not.

There is a clear positive trend, where higher parental education levels are associated with higher average scores. For example, students with higher parental education achieve average scores of 10.62 (no nursery) and 11.57 (nursery), compared to 9.56(no nursery) and 8.78(nursery) at the primary level.

Table 1 shows the exact values and sample sizes behind these results.

The effect of nursery attendance varies across education levels. For students with no parental education, the difference is large 14.67(Nursery) vs 8.00(No Nursery), but this is based on extremely small sample sizes by only 1 observations in both groups, so it should be treated with caution.

For the other categories, the effect is smaller and less consistent. For example, in the 5th-9th grade group, students without nursery attendance slightly outperform those with nursery 10.54(No Nursery) vs 10.04(Nursery), while in secondary education the opposite pattern is observed 11.50(Nursery) vs 9.12(No Nurserly).

Overall, parental education appears to be more consistently associated with academic performance than nursery attendance. This aligns with previous research linking socioeconomic background to academic performance [1]. While nursery attendance may provide some benefit, its impact seems limited and depends on the context.

3.4 Parental Education and Nursery Attendance (Portuguese)

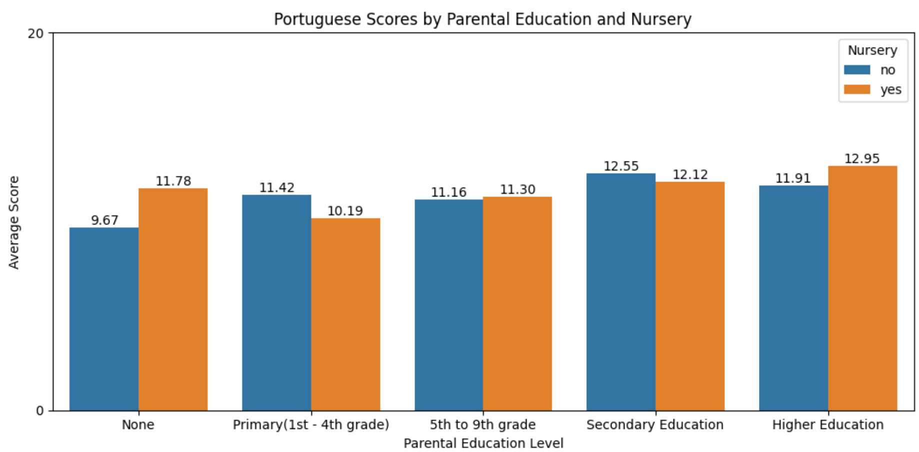

Fig. 4. Average Portuguese scores by parental education level and nursery attendance. Scores increase with parental education, while the effect of nursery attendance is smaller and less consistent.

Table 2. Average Portuguese Scores by Parental Education Level and Nursery Attendance with sample sizes

Nursery	None	Primary	5th–9th	Secondary	Higher
No	9.67 (n=2)	11.42 (n=23)	11.16 (n=69)	12.55 (n=11)	11.91 (n=22)
Yes	11.78 (n=6)	10.19 (n=71)	11.30 (n=223)	12.12 (n=77)	12.95 (n=135)

In Figure 4, Portuguese scores are compared across parental education levels, while separating students who attended nursery school from those who did not.

There is a clear trend where higher parental education is linked to the students higher scores. For example, students with higher parental education have average scores of 11.91 (No Nursery) and 12.95 (Nursery), compared to 11.42(No Nursery) and 10.19(Nursery) at the primary level.

Table 2 shows the exact values and sample sizes used in this comparison.

The effect of nursery school is generally small and not consistent across all the groups. In some cases, such as students with no parental education, nursery attendance is linked to higher scores 11.78(Nursery) vs 9.67(No Nursery). In other cases, such as primary education, the opposite can be seen 10.19(Nursery) vs 11.42(No Nursery).

Some of these differences are based on very small sample sizes, especially in the lowest parental education group (n = 2 and n = 6), so they should be interpreted carefully.

Overall, parental education seems to have a more stable relationship with performance, while the effect of nursery school is weaker and varies more depending on the groups.

This is similar to what was seen in the Mathematics dataset, which supports the idea that parental education plays a stronger role overall.

3.5 Family Size and Nursery Attendance

Table 3. Average Mathematics Scores by Family Size (GT3 = more than 3 members, LE3 = 3 or fewer) and Nursery Attendance (with sample sizes)

Nursery	**GT3**	**LE3**
No	10.14 (n=65)	10.43 (n=14)
Yes	10.50 (n=211)	11.33 (n=95)

In Table 3, family size is represented using the original dataset categories, where GT3 indicates households with more than three members, and LE3 indicates households with three or fewer members.

Table 4. Average Portuguese Scores by Family Size (GT3 = more than 3 members, LE3 = 3 or fewer) and Nursery Attendance (with sample sizes)

Nursery	**GT3**	**LE3**
No	11.49 (n=102)	11.20 (n=25)
Yes	11.58 (n=349)	12.00 (n=163)

In Table 4, family size is represented using the original dataset categories, where GT3 indicates households with more than three members, and LE3 indicates households with three or fewer members.

Family size (LE3 vs GT3) shows only small differences in academic performance across both datasets. In the Mathematics dataset (Table 3), the differences between family size groups are limited. For example, students from larger families (GT3) have average scores of 10.14(No Nursery) and 10.50(Nursery), compared to 10.43(No Nursery) and 11.33(Nursery) for smaller families (LE3). The largest difference is only 0.83 points, which suggests a relatively small effect.

A similar pattern can be seen in the Portuguese dataset (Table 4), where differences remain small across all the groups. For instance, within the nursery group, students from smaller families (LE3) have an average score of 12.00 compared to 11.58 for larger families (GT3), a difference of only 0.42 points.

Overall, the family size appears to have a weak and inconsistent relationship with academic performance/scores. Compared to factors such as parental education, the effect of family size seems somewhat limited.

It is also important to consider that the groups are uneven in size (like, n = 14 vs n = 211), meaning that some estimates are based on relatively small observations. This reduces reliability, so the observed differences should be treated with caution.

Across all the analyses, parental education stands out as the most consistent factor associated with student performance, showing a clear positive relationship in both Mathematics and Portuguese datasets. Nursery school shows a smaller but fairly consistent positive pattern, while family size has only a minor and inconsistent impact.

These findings suggest that the long term factors, such as family background, may have a stronger influence on academic outcomes than attendance of nursery school or family size alone. At the same time, the relatively small differences and uneven group sizes mean that the results should be interpreted carefully, and that student performance is likely influenced by several factors working together rather than a single variable.

3.6 Paid Extra Classes and Academic Performance (Mathematics)

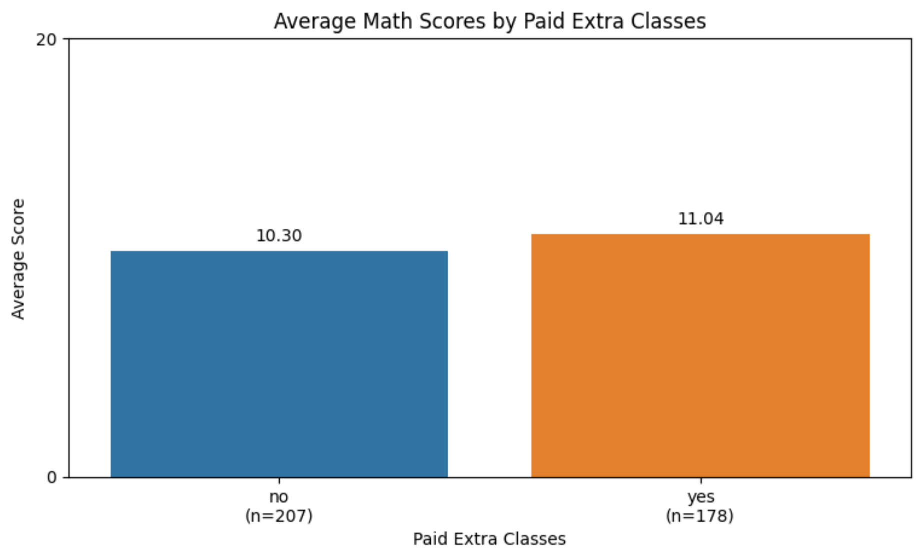

Fig. 5. Average Mathematics scores by participation in paid extra classes. A small difference is observed, with sample sizes.

In Figure 5, the average Mathematics scores are compared between students who receive paid extra classes and those who do not.

Students who receive paid extra classes have an average score of 11.04 (n = 178), compared to 10.30 (n = 207) for students who do not. This gives a difference of 0.74 points in favour of students receiving additional instruction.

This could suggest a positive relationship between paid extra classes and performance. However, the difference is relatively small, and does not necessarily imply a causal effect.

To test whether this difference is statistically significant, an independent two-sample t-test was performed.

The hypotheses for the test are:

$$H_0 : \mu_{\text{yes}} = \mu_{\text{no}}$$

$$H_1 : \mu_{\text{yes}} \neq \mu_{\text{no}}$$

The test statistic is calculated as:

$$t = \frac{\bar{x}_{\text{yes}} - \bar{x}_{\text{no}}}{\sqrt{\frac{s^2_{\text{yes}}}{n_{\text{yes}}} + \frac{s^2_{\text{no}}}{n_{\text{no}}}}}$$

where $\bar{x}$ is the sample mean, s represents the sample standard deviation, and n is the sample size for each group.

The p-value is calculated like this:

$$p = 2 \cdot P(T > |t|)$$

where T follows a t-distribution with the relevant degrees of freedom.

Using this approach, the test resulted in a t-value of 2.03 with approximately 383 degrees of freedom, and a p-value of 0.043.

At a significance level of $\alpha = 0.05$, the null hypothesis is rejected if $p < 0.05$.

Since the p-value 0.043 is below this limit, the null hypothesis was rejected. This means that there is evidence of a difference in average scores between the two groups.

However, even though the result is statistically significant, the practical impact appears limited. The difference in mean scores (0.74 points on a 0-20 scale) is relatively small.

It is also worth noting that the large sample size is incresing the sensitivity of the test, meaning that small differences can become statistically significant. Overall, this suggests that paid extra classes may be associated with slightly higher scores, but are unlikely to be a major factor compared to stronger influences such as parental education.

4 Student Profile Development

4.1 Variable Selection

Variables were selected based on their relationship with academic performance in both the Mathematics and Portuguese datasets. The goal was to focus on those that show clear patterns and can help describe different student profiles.

Table 5. Variables selected for student profile development based on correlation with Scores.

Variable	Math (r)	Portuguese (r)	Role	Reason
Failures	-0.37	-0.41	Strongest factor	Strongest negative relationship in both datasets
Parent education avg	0.21	0.29	Supporting factor	Represents family background influence
Study time	0.13	0.27	Supporting factor	Behavioural factor affecting performance
Medu	0.22	0.28	Excluded	Redundant with parent_edu_avg
Fedu	0.17	0.24	Excluded	Redundant with parent_edu_avg
Absences	-0.00	-0.13	Excluded	Weak and inconsistent relationship
Goout	-0.15	-0.07	Excluded	Weak negative relationship

Table 5 shows how the variables were selected. Correlation analysis was used to compare how each variable relates to the Score variable.

Failures is clearly the strongest factor, with a negative relationship in both Mathematics with $r = -0.37$ and Portuguese with $r = -0.41$. This suggests that students with more past failures generally achieve lower scores.

Parental education (using $parent\_edu\_avg$) and study time show moderate positive relationships in both datasets. These were kept as supporting factors, as they reflect both family backgrounds and study behaviour.

Other variables, such as absences and social activity (goout), do not show clear patterns and were not included. Variables that represent the same information, like Medu and Fedu, were combined into a single variable($parent\_edu\_avg$).

4.2 Student Profiles

Based on the exploratory analysis, three student profiles were identified using key variables such as past failures, parental education, and study time.

The first profile, *at-risk students*, consists of students with multiple failures in the past, lower parental education, and have lower study time. These students generally have the lowest scores across both subjects, which suggests that previous academic performance and support factors matter.

The second profile, *high-performing students*, includes students with no or few failures, higher parental education, and higher study time. These students tend to achieve the highest scores and show more stable performance overall.

The third profile, *moderate students*, includes students with relatively stable backgrounds, moderate study habits, and few failures. Their performance is consistent, but still below the highest performing students group.

Overall, past failures appear to be the strongest factor, while parental education and study time also contribute and help explain differences between students.

4.3 Failures and Study Time

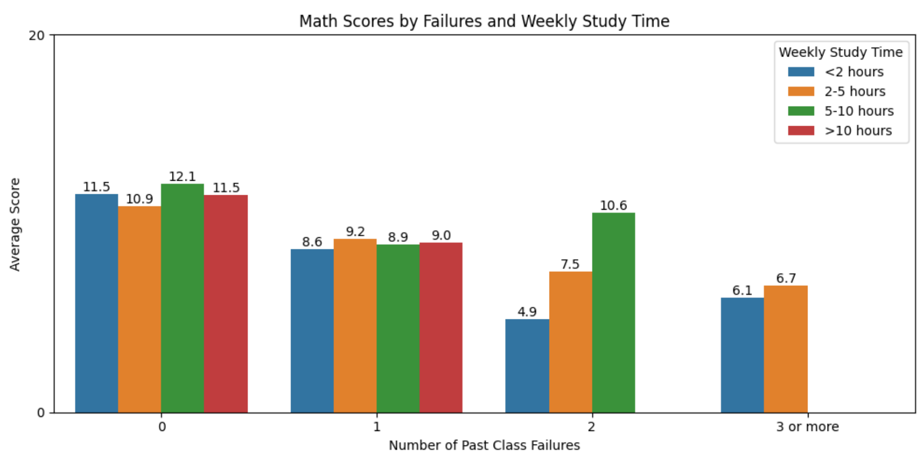

Fig. 6. Average Mathematics scores by number of past class failures and weekly study time. Scores decrease as failures increase, while higher study time corresponds to higher scores in most groups.

Table 6. Sample Sizes for Mathematics by Number of Past Failures and Weekly Study Time

Failures	**<2h**	**2-5h**	**5-10h**	**>10h**
0	71	155	53	25
1	16	24	7	1
2	6	7	4	–
3+	9	7	–	–

Figure 6 shows the relationship between past class failures and weekly study time in the Mathematics dataset. There is a clear negative trend, where average scores decrease as the number of failures increases. For example, students with no failures have average scores around 11, while those with two or more failures drop to much lower levels around 4.9-6.7 depending on the study time.

At the same time, higher study time appears to be linked to better performance within most failure categories. For instance, among students with two failures, average scores increase from 4.9 for low study time to 10.6 for moderate study time(5-10 hours per week). This means that higher study time can partly make up for earlier poor performance in some cases.

Table 6 shows the corresponding sample sizes. Some subgroups, especially those with higher failure levels and higher study time, contains relatively few observations (like, n = 4 or n = 1). This means the estimates for these groups are less reliable and should be interpreted with care.

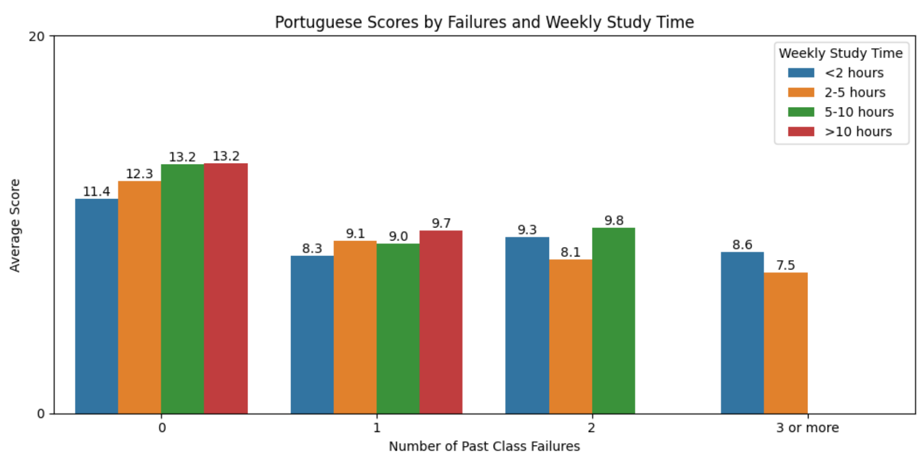

Fig. 7. Average Portuguese scores by number of past class failures and weekly study time. Scores decrease as failures increase, while higher study time corresponds to higher scores in most groups.

Table 7. Sample Sizes for Portuguese by Number of Past Failures and Weekly Study Time

Failures	<2h	2–5h	5–10h	>10h
0	159	261	88	32
1	36	25	6	2
2	4	10	2	–
3+	9	5	–	–

A similar pattern can be seen in the Portuguese dataset, as shown in Figure 7. Academic performance again declines as the number of failures increases, which shows that this relationship is consistent across both subjects. For example, average scores decrease from around 12.2 (no failures) to about 8.2 for students with three or more failures.

Higher study time also appears to be linked to better performance within most groups. For example, among students with one failure, average scores increase from 8.3 to 9.7 as study time increases. This suggests that study behaviour may help improve the academic performance.

However, as shown in Table 7, sample sizes become much smaller for higher failure categories. Some combinations include very few observations, which means the averages are less reliable and should be interpreted with care.

4.4 Failures and Parental Education

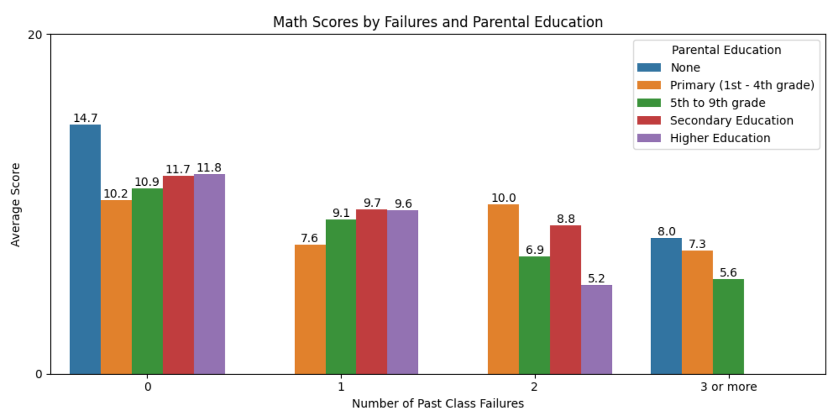

Fig. 8. Average Mathematics scores by number of past class failures and parental education level. Scores decrease as failures increase, while higher parental education corresponds to higher scores in most groups.

Figure 8 shows the relationship between the number of past class failures and parental education level in the Mathematics dataset. There is a clear negative trend, where average scores decrease as the number of failures increases. For example, students with no failures have average scores ranging from about 10.2 to 11.8 across most parental education levels, while students with two or more failures have much lower scores, in some cases dropping below 6.

At the same time, higher parental education appears to be linked to better performance, especially for students with no or few failures. For instance, students with higher parental education have an average score of 11.8 (no failures), compared to 10.2 for students with primary level parental education. This suggests that family background might play a role in academic performance outcomes.

However, the effect of parental education becomes less consistent at higher failure levels. For example, in the group with two failures, some categories show irregular patterns, which may be influenced by smaller sample sizes.

Table 8. Sample Sizes for Mathematics by Number of Past Failures and Parental Education Level

Failures	**None**	**Primary**	**5–9th**	**Secondary**	**Higher**
0	1	21	119	50	113
1	–	10	23	9	6
2	–	2	7	4	4
3+	1	6	9	–	–

Table 8 highlights this issue. Several subgroups have very few observations (like, n = 1 or n = 2), particularly in the “None” category and among students with more failures. Because of this, the estimates for these groups are less reliable and need to be treated with caution.

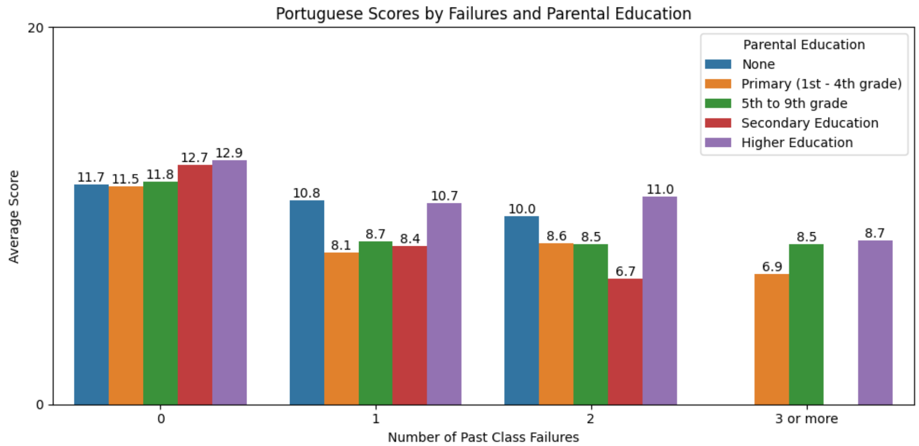

Fig. 9. Average Portuguese scores by number of past class failures and parental education level. Scores decrease as failures increase, while higher parental education corresponds to higher scores in most groups.

Table 9. Sample Sizes for Portuguese by Number of Past Failures and Parental Education Level

Failures	None	Primary	5-9th	Secondary	Higher
0	5	66	243	78	148
1	2	19	32	9	7
2	1	6	7	1	1
3+	–	3	10	–	1

A similar pattern can be seen in the Portuguese dataset. As shown in Figure 9, academic performance decreases as the number of failures increases.

This shows a clear negative relationship between failures and scores across both subjects. Higher parental education also appears to be linked to better performance across most groups.

For example, students with higher parental education have an average score of 12.9 in the no failure group, compared to 11.5 for those with primary level parental education.

However, as shown in Table 9, sample sizes become smaller at higher failure levels. Several combinations, especially those with higher failures and lower parental education, include very few observations. This means the averages for these groups are less reliable and should be interpreted with care.

Overall, parental education appears to have a positive and a consistent relationship with performance, particularly at lower failure levels. At the same time, past class failures seem to have a stronger influence overall, while parental education plays a supporting role.

4.5 Comparison Across Subjects

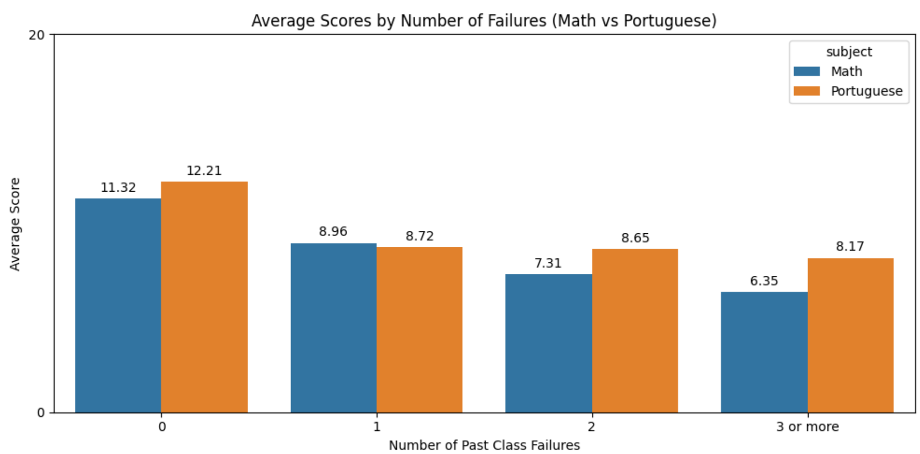

Fig. 10. Average scores in Mathematics and Portuguese by number of past class failures. Scores decrease as failures increase in both subjects, while Portuguese scores are slightly higher in most groups.

Figure 10 compares average scores in Mathematics and Portuguese across different levels of past class failures. Scores decrease in both subjects as failures increase. Students with no prior failures have an average scores of 11.32 in Mathematics and 12.21 in Portuguese, while those with three or more failures drop to 6.35 and 8.17.

This suggests a strong relationship between past failures and performance in both datasets. Portuguese scores are generally slightly higher, but the overall pattern is very similar, which indicates that the effect of failures is consistent across the two subjects.

Overall, past academic performance seems to be one of the main variables explaining differences in student academic outcome.

5 Regression Modeling

5.1 Model Setup

Multiple linear regression was used to model and predict the students Mathematics scores. The target variable was *Scores*, and several sets of predictors/variables were tested to see how they affect the model performances.

The baseline model included only the number of past class failures, as this showed the strongest relationship with performance in the exploratory analysis.

The core model then added study time and parental education ($parent\_edu\_avg$), capturing both study behaviour and family educational backgrounds.

A extended model was also tested, adding variables like absences, social activity (goout), and health, to check whether more variables would improve the results.

The dataset was split into training and test sets using an 80/20 split. This makes it possible to evaluate the model on unseen data and get a better sense of its generalization ability.

5.2 Model Evaluation

Table 10. Model Performance Comparison

Model	RMSE	R^2
Baseline	3.80	0.09
Core	3.75	0.11
Extended	3.68	0.14
Core + higher	3.78	0.10
Polynomial (deg 2)	3.74	0.12
Polynomial (deg 3)	3.73	0.12

Looking at Table 10 it shows that all models have low R^2 values, which means that only a small part of the variation in student scores are explained.

Model performance was evaluated using RMSE and R^2, giving a view of both prediction accuracy and explanatory power.

The baseline model has an RMSE of 3.80 and an R^2 value of 0.09. This suggests that using failures alone gives a limited predictive performance.

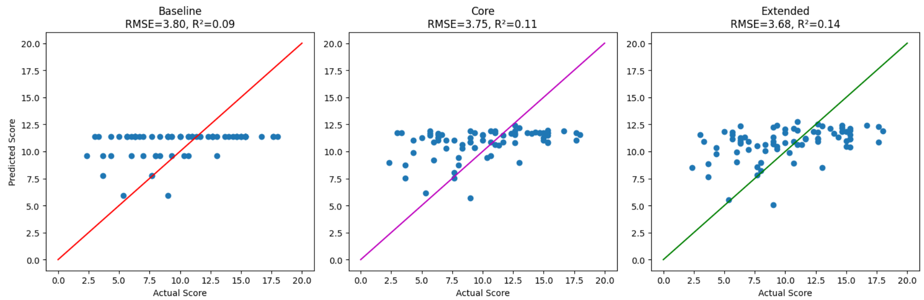

Fig. 11. Comparison of baseline, core, and extended regression models. Each plot shows actual vs predicted scores, where the diagonal line represents perfect predictions. The clustering of predictions around a narrow range indicates limited model performance and underfitting.

Figure 11 shows that the baseline model produces predictions that are close together and do not follow the diagonal line of perfect predictions. This means that the model captures very little of the variation in the data and tends to predict values close to the mean, which matches the low R^2 values in Table 10.

The core model performs slightly better with RMSE $= 3.75$ and $R^2 = 0.11$. This suggests that study time and parental education contribute additional information to the model. This is also visible in Figure 11, where the predictions are a bit more spread out.

The extended model gives the best results, with $R^2 = 0.14$ and RMSE $=$ 3.68. Even so, the improvement is small the predictions are still concentrated around the mean. This suggests that adding more variables does not significantly improve the model.

5.3 Model Comparison

Polynomial regression models of degree 2 and 3 were tested using the core feature set to see how model complexity affects performance.

Predicted versus actual values were plotted for both linear and polynomial models to compare the effect of increasing complexity.

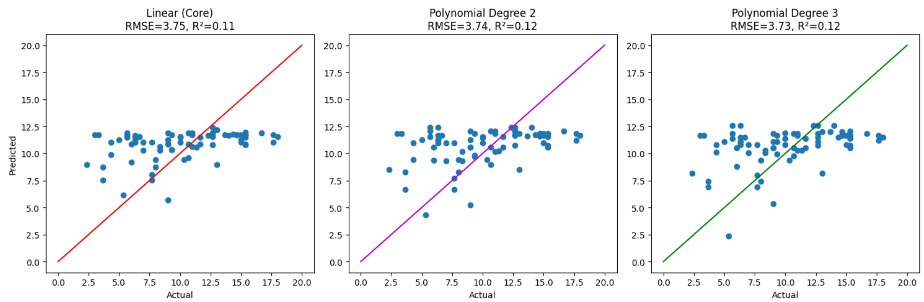

Fig. 12. Comparison of linear and polynomial regression models. Increasing model complexity gives only a small improvement, suggesting that the relationships in the data are weak or close to linear.

The degree 2 model has RMSE $= 3.74$ and $R^2 = 0.12$, while the degree 3 model has RMSE $= 3.73$ and $R^2 = 0.12$. The improvement compared to the linear model is very small.

Figure 12 shows that Increasing model complexity by adding more variables does not make predictions closer to the actual values. Predictions remain concentrated within a narrow range across all models, which means the models capture only a small amount of variation.

This suggests that the relationship between the selected variables and student performance is not strongly non linear. Increasing model complexity therefore gives little improvement in the predictive performance.

The scatter plots comparing actual and predicted values show a similar pattern. Predictions tend to cluster around the mean, which suggests that the models struggle to capture variation in the data. This is consistent with underfitting, where the model does not capture enough of the underlying patterns in the data.

To further explore the impact of additional variables, a binary variable representing students intention to pursue higher education (*higher*) was tested.

Including this variable, it slightly reduced the model performance (RMSE $=$ 3.78, $R^2 = 0.10$) compared to the core model ($R^2 = 0.11$). This suggests that, while the variable may reflect group differences, it does not add much predictive value when combined with the other features.

Overall, these results suggest that adding more variables does not necessarily improve the model, and that only a small set of predictors contributes meaningful explanatory value in this dataset.

5.4 Final Model and Interpretation

Based on the evaluation, the core model was selected as the final model for the predictions. Even though the extended model shows slightly better results with RMSE = 3.68, $R^2 = 0.14$, but the improvement compared to the core model (RMSE = 3.75, $R^2 = 0.11$) is small. The core model was therefore chosen, since it is simpler, easier to interpret, and based on variables that showed clear relationships in the exploratory analysis.

Additional testing supports this choice further. When the binary variable *higher* was included, model performance decreased ($R^2 = 0.10$). This suggests that not all the variables that appear relevant actually improves the model.

The final model is based on multiple linear regression and can be written like this:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n$$

where y is the predicted outcome, β_0 is the intercept, and β_i are the coefficients for each predictor.

For this case, the fitted model goes as follows:

$$\text{Scores} = 9.40 - 1.63 \cdot \text{failures} + 0.28 \cdot \text{studytime} + 0.47 \cdot \text{parent\_edu\_avg}$$

The coefficient for failures with -1.63 shows a strong negative relationship, meaning that each additional failure lowers the predicted score with around 1.63 points.

Study time with $+0.28$ and parental education with $+0.47$ shows positive relationships, but their impact is smaller compared to the impact of failures.

Even with these relationships, the overall model performance is still low with only $R^2 = 0.11$, which means that only a small part of the variation in student scores is actually explained by the model.

Overall, the model follows the same pattern as the exploratory analysis, where failures have the strongest impact. At the same time, it still leaves a lot of the differences in student scores unexplained.

5.5 Model Application and Predictions

Table 11. Predicted Mathematics Scores for Prediction Sample

Student ID	Predicted Score
1	11.87
2	10.61
3	9.26
4	11.54
5	11.82
6	10.61
7	9.26
8	11.59
9	11.36
10	12.15

The final regression model was applied to the 10 students in the Prediction_Sample dataset using the selected predictors: failures, study time, and parental education average.

A `StudentID` variable was added to identify each observation and connect the input data to the predicted values, as shown in Table 11.

The predicted Mathematics scores are presented in Table 11, keeping the original order so each prediction matches the corresponding input.

The predicted scores have much lower variability than the actual scores, when looking at the standard deviation of the predictions with 1.05, compared to 3.69 for the full Mathematics dataset.

This suggests that the model tends to predict values closer to the mean, which matches the underfitting we have seen earlier. The model captures the overall level of performance, but not the full spread of the data, as shown by the narrow prediction range (9.26 to 12.15) in Table 11.

6 Classification Modeling

6.1 Target Variable Construction

The regression problem was converted into a classification task by defining a binary variable, *Proficiency*, based on the students Portuguese scores.

The dataset has an average score of 11.66, students with scores at or above this value are labelled as *Good*, while those below are labelled as *Not Good*.

This makes it possible to treat the outcome as a binary classification problem and apply classification models.

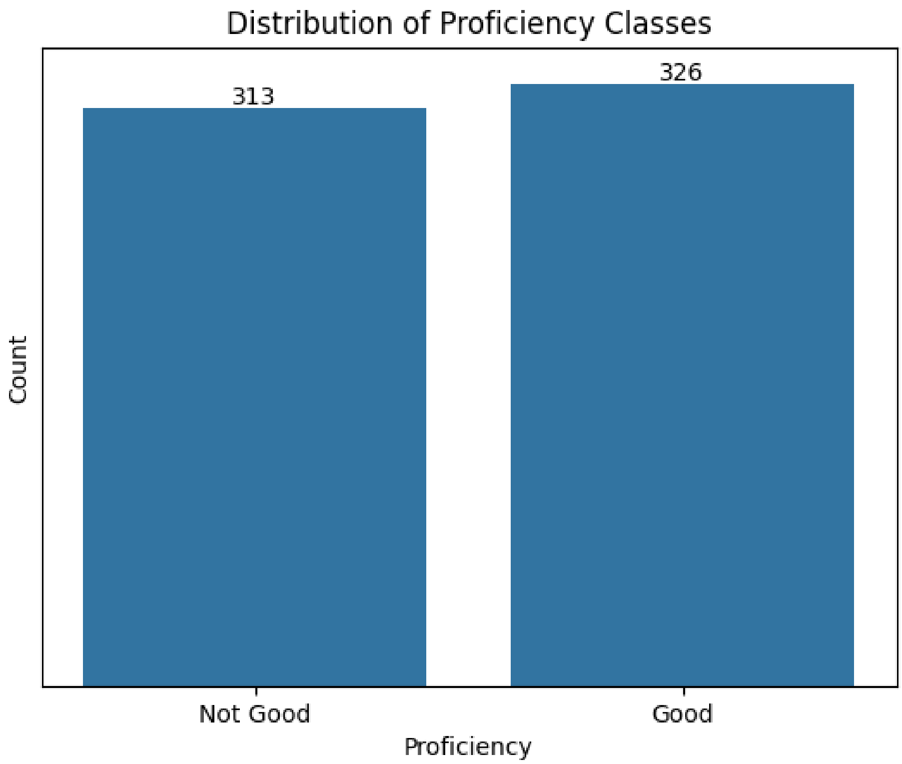

Fig. 13. Distribution of proficiency classes. The data is close to evenly split between *Good* and *Not Good* students.

Figure 13 shows a balanced class distribution, with 326 observations labelled as *Good* and 313 as *Not Good*. This suggests that class imbalance is not a major issue in this dataset.

6.2 Model Setup

A logistic regression model was used to classify students into *Good* and *Not Good* categories.

The dataset was divided into a feature matrix (X) and a target variable (y), where y represents the Proficiency classification.

To avoid data leakage, *Scores* and *Proficiency* were excluded to avoid data leakage, together with variables with overlapping information (like *Medu* and *Fedu*) were also removed, keeping only *parent_ edu_ avg*.

Categorical variables were converted into numerical values using simple encoding methods and the data was also standardized to keep the model stable.

The dataset was split into training and test sets using an 80/20 split, making it possible to evaluate the model on unseen data.

6.3 Model Evaluation

The model achieved an accuracy of 70.31% on the test dataset, this indicates a moderate classification performance since it is below 90% and above 60%.

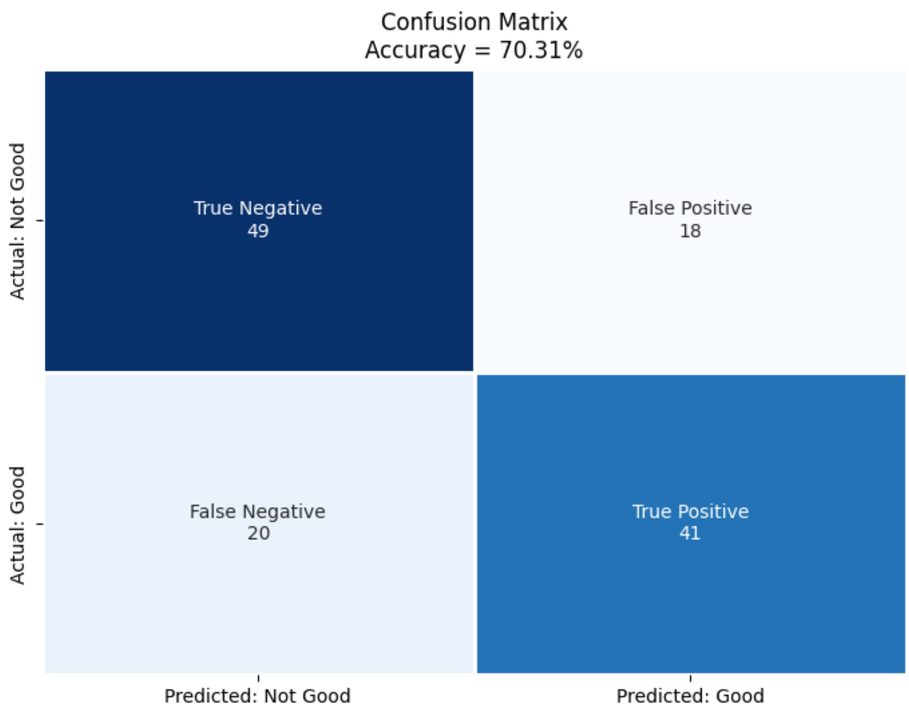

Fig. 14. Confusion matrix for the classification model. Most observations are classified correctly, but some false positives and false negatives are present.

Figure 14 shows the classification results. The model correctly classifies 49 *Not Good* students (true negatives) and 41 *Good* students (true positives). At the same time, 18 *Not Good* students are classified as *Good* (false positives), and 20 *Good* students are classified as *Not Good* (false negatives).

The classification report gives a more detailed view of performance:

- **Good:** Precision = 0.71, Recall = 0.73, F1-score = 0.72
- **Not Good:** Precision = 0.69, Recall = 0.67, F1-score = 0.68

Precision shows how many of the predicted positive cases are correct:

$$\text{Precision} = \frac{TP}{TP + FP}$$

Recall shows how many of the actual positive cases the model correctly identifies:

$$\text{Recall} = \frac{TP}{TP + FN}$$

The F1-score looks at both precision and recall at the same time:

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

For the *Good* class, the recall is 0.73, meaning that 73% of the actual *Good* students are correctly identified. For the *Not Good* class, the recall is lower (0.67), which suggests that the model misses more of these students.

The F1-scores (0.72 for *Good* and 0.68 for *Not Good*) follow the same pattern, with slightly better performance for the *Good* class.

Overall, the model performs reasonably well, but it appears to favor the *Good* class. This means that some lower performing students are more likely to be misclassified.

6.4 Feature Influence

The model was interpreted by examining the coefficients of the most influential features.

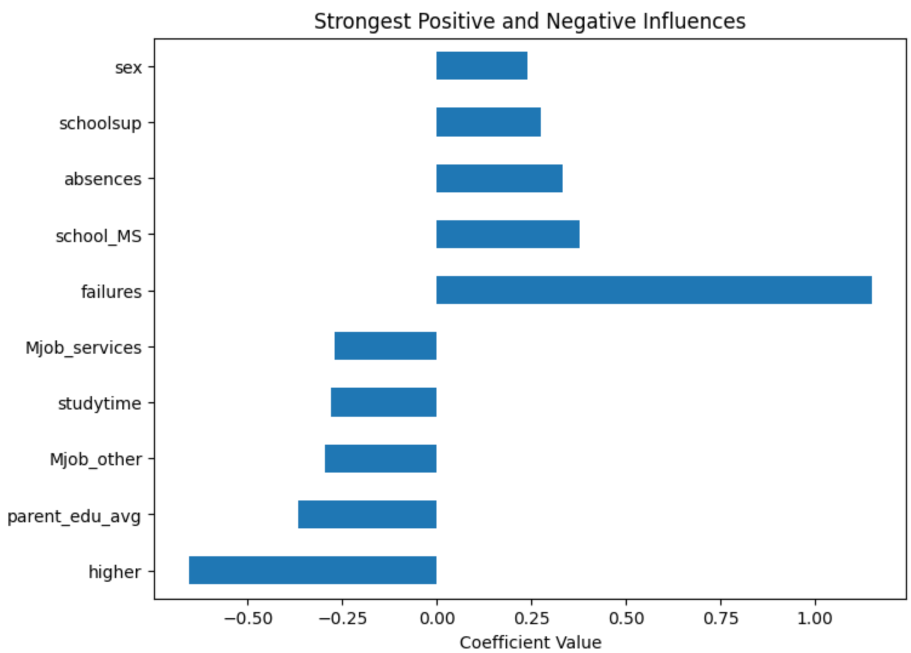

Fig. 15. Feature influences in the logistic regression model. Positive and negative effects are shown for the strongest variables.

Figure 15 shows that past failures have the strongest negative impact on student proficiency. Students with more failures are therefore less likely to be classified as *Good*.

Study time and parental education have smaller positive effects, while variables such as higher education intention show a negative effect in this model.

6.5 Model Performance and Interpretation

The classification model achieves an accuracy of 70.31% on the test set, meaning that 90 out of 128 students are correctly classified. This corresponds to moderate predictive performance.

The evaluation is based on the test set (20% of the data), so the results reflect how the model performs on unseen data.

The confusion matrix shows some clear limitations. The model correctly classifies 49 *Not Good* students and 41 *Good* students. At the same time, 18 *Not*

Good students are classified as *Good* (false positives), and 20 *Good* students are classified as *Not Good* (false negatives).

This means that the model has difficulty separating the two groups, for example, predicting a *Not Good* student as *Good* may lead to overestimating performance, while predicting a *Good* student as *Not Good* does the opposite.

The class distribution is balanced (326 *Good* vs. 313 *Not Good*), but the model still performs slightly better for the *Good* class. This can be seen in the recall values, where *Good* has 0.73 compared to 0.67 for *Not Good*.

This is similar to what we saw in the regression analysis, where the weak relationships between variables made it hard for the model to make accurate predictions. The model captures general trends, but it does not clearly separate students at the individual levels.

Overall, the model captures general patterns, but its predictive accuracy is limited, especially at the individual level.

7 Conclusion

This report looked at student performance using an exploratory analysis, regression modeling, and classification modeling across both the Mathematics and Portuguese datasets.

From the exploratory analysis, one thing that stands out is that the past class failures have the strongest relationship with performance/scores. The correlations are -0.37 for Mathematics and -0.41 for Portuguese, which are noticeably stronger than the other variables found in the datasets. Parental education shows a positive relationship 0.21 in Mathematics and 0.29 in Portuguese, and the study time also seems to matter, especially when looking at students who already have some academic challenges. However, Other variables like nursery school attendance and family size, show a much smaller or inconsistent impact.

This generally fits with what is discussed in the literature, where family background is often linked to academic outcomes and performance [1]. Parental education in particular likely reflects access to for example more support, resources, and learning conditions outside the classroom.

When moving to modelling, the results are much weaker. The regression model, which includes failures, study time, and parental education, follows the expected pattern, but the overall performance is still quite low with $R^2 = 0.11$, the model explains only a small part of the variation in the student scores.

The classification model shows similar limitations, it reaches an accuracy of only 70.31%, with 90 out of 128 students correctly classified. However, the performance of the model is not poor, but the confusion matrix shows that the model is still struggling to separate students near the boundary between the two groups.

Taken together, the results point in the same direction, with Failures seem to matter the most, and parental education also plays a role, but the overall relationships are not strong enough to support highly accurate predictions for the models.

Because of this, the results are better understood as general patterns rather than precise predictions. It is likely that student performance depends on a wider set of factors that are not fully captured in this dataset [1].

References

1. Sirin, S.R.: Socioeconomic Status and Academic Achievement: A Meta-Analytic Review of Research. Review of Educational Research **75**(3), 417–453 (2005). https://doi.org/10.3102/00346543075003417, https://journals.sagepub.com/doi/10.3102/00346543075003417